

EECE 571Q

Lecture 14: Decoding

Tuesday 14 July 2026

Lecture Outline

- **Review of decoding objective**
- **Classical decoding algorithms**
 - Union Find
 - MWPM and the Blossom algorithm
 - Belief Propagation w/ OSD and LSD
- **Real Time Decoding**
 - Requirements for RT decoding
 - Decoder error models
 - Performance metrics
- **Demonstration**

Learning Outcomes

- **LO1:** Formulate quantum decoding as an optimization problem over recovery operations
- **LO2:** Explain why maximum-likelihood decoding (MLD) is generally impractical and approximate decoders are needed
- **LO3:** Apply the algorithms of Belief Propagation (BP) and its variants, Minimum Weight Perfect Matching (MWPM) and Union-Find (UF), from syndrome to recovery
- **LO4:** Identify common performance metrics in QEC decoding, comparing algorithms and adapting to face the challenge of real-time decoding

Last time

- Tanner Graphs
- Belief Propagation + Ordered Statistical Decoding
- Minimum Weight Perfect Matching

$$\underset{\text{Decoder}}{D(s)} \mapsto \underset{\text{Recovery}}{R}$$

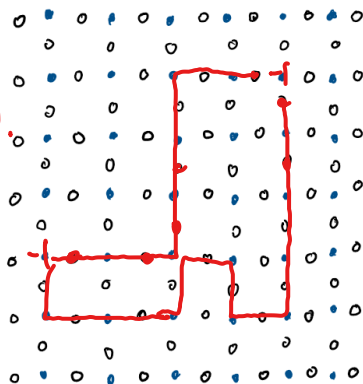
Maximum likelihood decoding

Choosing error with the highest probability of occurring:

$$S \mapsto R.$$

$$L_* = \underset{L}{\operatorname{argmax}} (P(L|S)).$$

$$s = He.$$



Many valid error chains.

$$P \in \{I, X, Y, Z\}.$$

4^{d^2} possible errors.

Sum over all satisfying syndrome.

What can we do?

Surrogate functions!

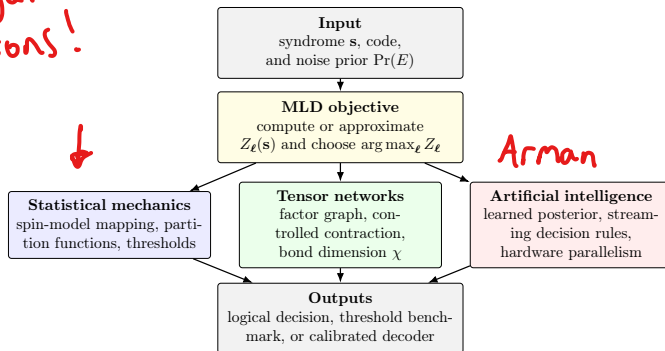
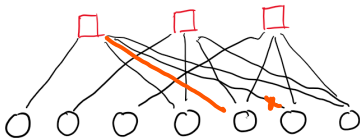
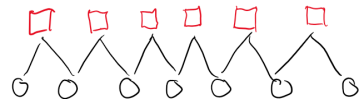


Figure: A classification of the recent approaches to quantum decoding adapted from [Cao, Yan, Du, Pan, 2026]

Tanner Graphs



Corrected
edge on
hamming code.

Sparsity makes our
lives easy.

- Less overhead
- Less degeneracy
- Easier Decoding.

QLDPC codes have bounded
edge weight.

Union-Find

Finds paired error syndromes on topological codes.

Steps:

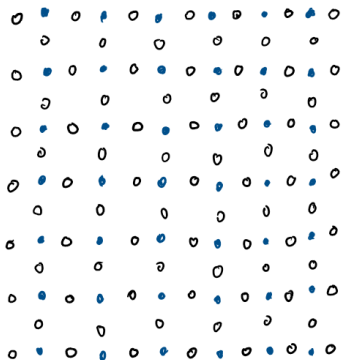
- ① Initialize clusters at syndrome defect locations
- ② Grow clusters outward by 1/2 edge at each step
- ③ Check parity, odd clusters have an unresolved error chain
- ④ Apply peeling algorithm finding a path between clusters and syndrome nodes



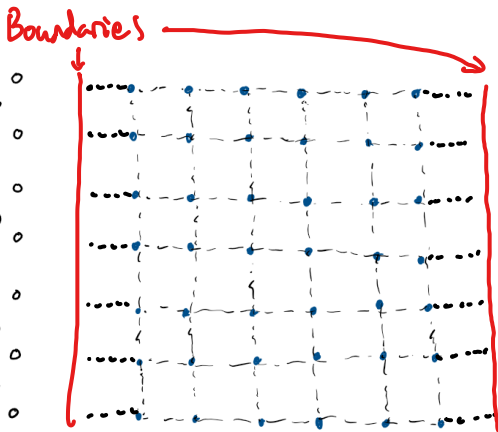
Will give a perfect matching, not necessarily minimum weight

Cluster connecting both boundaries can have a logical error

UF Example

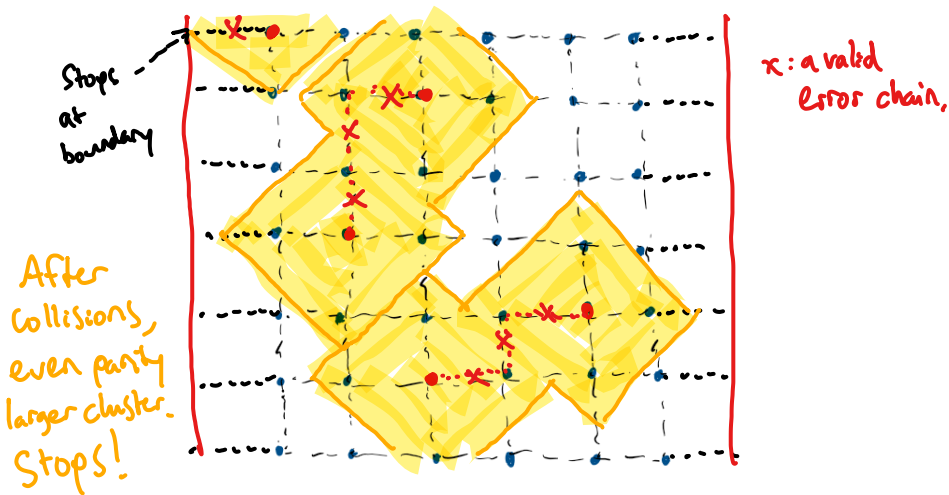


$d = 7$ Surface Code w/ Z
stabilizers



Union Find Setup

UF Example Ctd.



MWPM and the Blossom algorithm

Dual-primal optimization problem.

Finds minimum weight paired error syndromes on topological codes.

$$e = \underset{e: H e = s}{\operatorname{argmin}} w(e)$$

$$w(e) = y_a + y_b + \dots$$

Steps:

- 1 Same setup as UF
- 2 If blossoms merge into odd cluster, begin a new blossom
- 3 Matches stop expanding
- 4 If another blossom merges with match, apply alternating tree
- 5 Blossoms terminate when even parity or border touched
- 6 Matching can only occur within the same blossom



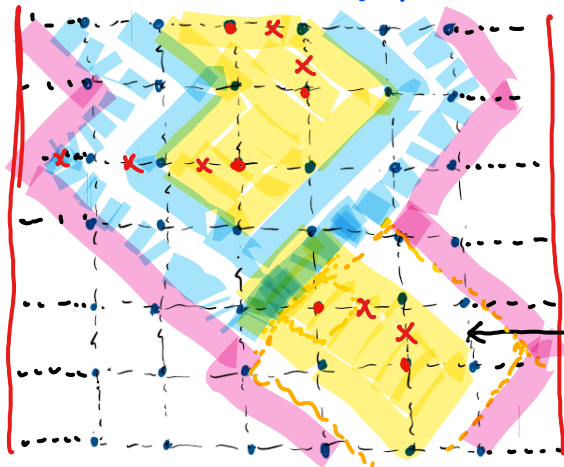
MWPM Example

Boundaries.

2. blossom

1. clusters
expand until
collision

3. Odd
Cluster
continues
expanding.
x's : valid
correction.

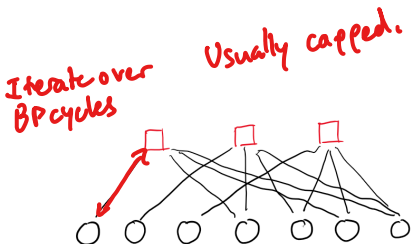


apply dual
Shrink and
growth on
Collision

unrotated $d=7$ surface

blossom expands
to space.

Belief Propagation (review)



IF we don't converge after t cycles, OSD

BP Steps

- 1 Initialize data bits with p
- 2 Error probability on physical bits passed to check nodes
- 3 Beliefs updated using syndrome and sum-product rule
- 4 Beliefs passed back to the data bits
- 5 Hard decision made on updated beliefs, check if $H_{eBP} = s$

OSD (review)

BP
Output

	0.2	0.1	-4	3	3	2	0.2
1	0	0	1	1	0	1	
0	1	0	1	0	1	1	
0	0	1	0	1	1	1	

Reorder by reliability

OSD Steps

- 1 Sort and reorder matrix by BP reliability
- 2 Gaussian Elimination to get pivots and free set
- 3 Solve for pivot bits by assigning hard decisions to free bits
- 4 Enumerate and score bit flip combinations in free set
- 5 Output lowest cost candidate error pattern

BP and Localized Statistics Decoding

OSD's major bottleneck is the global matrix solve

We need a way of limiting this overhead by parallelizing this step

Idea - Only solve local parity check matrices using OSD, treat as clusters

Then we can parallelize each cluster

LSD: Local Matrix Solves

Global decoding problem: $He = s$

LSD separates this into local clusters which can be solved in parallel

$$H_{C_i} e_{C_i} = s_{C_i}$$

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} e_3 \\ e_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

LSD: On-the-Fly Elimination

1. Initial C_1

$$\left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{array} \right] \xrightarrow{+R_1+R_2} \begin{array}{c} e_1 \quad e_2 \\ \left[\begin{array}{cc|c|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array} \right] \end{array}$$

invalid.

2. Grow C_1

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{array} \right] \xrightarrow{+R_1+R_2} \begin{array}{c} \left[\begin{array}{ccc|c|c} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{array} \right] \end{array}$$

$$e_1 = 1, \quad e_2 = 0, \quad e_3 = 1$$

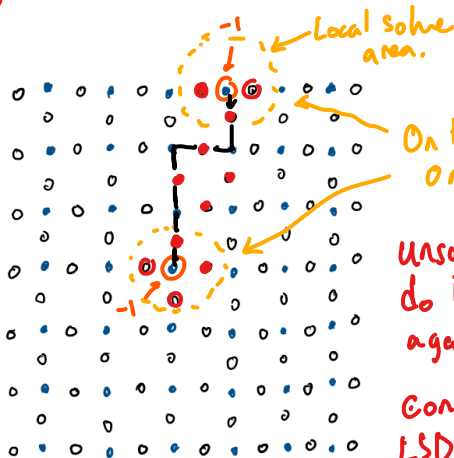
Gaussian Elim
on much smaller
matrices, especially
on larger codes.

And parallel
:)

Localized Statistics Decoding

● high prob
○ low prob. $> BP$

Final
connected
cluster.
Satisfies
LSD eqn.



On the fly elim
on each cluster.

unsolved so we
do BP and expand
again.

Continue until
LSD matrix solvable

Interlude

Questions?

All references will be included in lecture notes, as well as some additional readings on some modern approaches to these algorithms such as fusion and sparse blossoms for MWPM

Requirements for real time decoding

If $r_{\text{gen}}/r_{\text{proc}} = f > 1$, a small initial backlog in processing syndrome data will lead to an exponential slow down during the computation. [Terhal, 2015]

$$\Delta_{\text{gen}} = f \Delta_{\text{proc.}}$$

$$D_1: \Delta_{\text{gen}} = f \Delta_{\text{proc.}}$$

$$D_2: \Delta_{\text{gen}} = f(f \Delta_{\text{proc.}})$$

$$D_N: \Delta_{\text{gen}} = f^N(\Delta_{\text{proc.}}) \quad \text{"}$$

$\approx 1 \mu\text{s}$ required for trapped ion systems.

Detector Error Models

Circuit / Hardware



Detector Error Model (DEM)



Decoder

- Bridge circuit level noise and decoder operation
- Probabilities associated with combinations of physical faults
- Encodes all this information and passes to decoder

Performance Metrics

Metrics used to evaluate QEC Decoders in practice:

- Error threshold t
- Logical Error rate p_l
- Decoding Time (per cycle) T_d

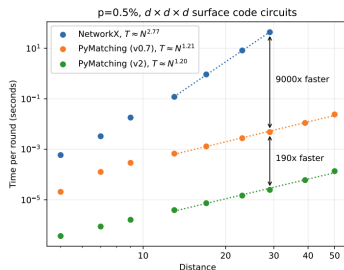


Figure: A plot of decoding time per cycle against distance for pymatching [Sparse Blossom: correcting a million errors per core second with minimum-weight matching, Higgott, Gidney, 2025]

Demo 1

Comparison of various decoders on different codes

Showcasing performance and capabilities of different decoders in different situations